Dylan He

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- **Birthday:** 2005.06.11
- **Phone:** 13608270773
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Research Interest / Position:
Physics / Mechanism-Informed AI for Clinical Insight

Education

2023 – **B.S., Sichuan University** Industrial Engineering.
GPA: 3.85 (8/73)
Major Courses: Calculus, Undergraduate Physics, Linear Algebra, Differential Equations, Probability Theory, Statistical Inference, Operations Research, and Simulation Modeling. Also Information System Engineering, Human Factor Engineering, Electrc Circuit, Facility Layout and Material Handling, Engineering Economic Anlysis.

Sichuan University Pittsburgh Institute BioFluid Lab.
Undergraduate Researcher

Research Publications

Journal Paper

1. S. Cheng, L. Liu, Y. Qiu, Y. Diao, **Daiyang He**, F. Yan, C. Wang, and K. Li, "Hypoxia promotes blood blister-like aneurysms development: A computational fluid dynamics study," *Journal of Biomechanics*, 2025.
2. S. Cheng, Y. Qiu, **Daiyang He**, M. L. Wang, C. Weng, D. Yuan, and K. Li, "Helical vortex arising from aberrant right subclavian artery induces occurrence of dissection tears," *Physics of Fluids*, 2025.
3. **Daiyang He**, C. Weng, S. Cheng, D. Yuan, T. Wang, W. Zeng, Y. Luo, J. Wang, K. Li, and Y. Qiu, "Noise-pressure constrained liutex method for robust vortex identification in 4d flow mri of abdominal aortic aneurysms," *Computer Methods and Programs in Biomedicine (under review)*, 2025.

Academic Experience

Awards and Achievements

2024 **Third Prize**, the Southwest Division of the China Undergraduate Physics Tournament (CUPT).

Third Prize, National Collegiate Mathematical Contest in Modeling.

S Award, American Collegiate Mathematical Contest in Modeling.

Project Experience

2023–2024 **Undergraduate Training Program (National Level)** "ShadowSift: An Efficient Real-Time Detection System for Low-Quality Deepfake Videos."
Responsible for the construction of dataset, model design, and core algorithm development.

2024 **National Natural Science Foundation of China (NSFC)** "Prediction of Adverse Events Related to Anchoring Area after Endovascular Treatment of Abdominal Aortic Aneurysm Based on Geometric and Biomechanical Analysis."
Responsible for the design of vortex identification algorithm based on 4D Flow MRI data to study the blood flow pattern in abdominal aortic aneurysm. (in progress)

Research Project "Study on the Biomechanical Mechanism of Dissection in Subclavian Vagus Artery Patients Based on Computational Fluid Dynamics."
Resbonsible for CFD Simulation, analysis of clinical CT images, and formal data analysis.

Academic Experience (continued)

2025 — **National Natural Science Foundation of China (NSFC)** "Multi-scale hemodynamic seepage characteristics of liver for non-invasive diagnosis of portal hypertension in liver cirrhosis."
Responsble for developing machine learning algorithm for prediction of clinical conditions of portal hypertension patients.(in progress)

Skills

- Languages — Strong reading, writing and speaking competencies for English.
- Coding — Python, C/C++, MatLab, Mathematica, NetLogo, LaTeX, SQL, …
- Science — Expertise in mathematics and science courses: Applied Functional Analysis, Complex Analysis, Statistical Inference, Probability Theory, Fluid Mechanics, Mchine Learning, Data Analysis, Medical Image Anslysis, …
- Software — Mimics, 3D Slicer, ANASYS Fluent, ICEM CFD, RadiAntViewer, ParaView, TecPlot, COMSOL, Tracker, SPSS, Microsoft Office, Canva, …

Miscellaneous Experience

- 2023–2024 — **Excellent Student**, Sichuan University.
- **University-level Comprehensive Third-class Scholarship**, Sichuan University.
- 2024 — **Excellent Leader**, Sichuan University "Students' Hometown Tour" practical activity.
- **Second Prize**, Biomedical Engineering Science Popularization Micro Video Competition
- 2024–2025 — **Outstanding Student Leader**, Sichuan University.
- **University-level Comprehensive Third-class Scholarship**, Sichuan University.

Student Organization

- Technology and Science Association — **Leader** of Technology and Science Association (TSA).
- SCUPI Student Council — Planning Department, Officer
- Class Council — IE-1 Class, Commissary in charge of studies and organization.
- Peer Advising Program — Peer Advisor.
- SCUPI Teaching Assistant — TA of 2025 Fall Productivity Analysis, 2026 Spring Information SYstem Engineering, Interdisciplinary Research.

Self-evalustion

My research interest lies in medical AI, specifically leveraging structural, geometric, and mechanistic priors to advance disease analysis and medical AI modeling. My undergraduate work spanned medical image analysis and image-based hemodynamic simulation; this combination of data-driven and physics-based perspectives constitutes my distinctive edge in developing physics-informed, clinically grounded AI systems.

Highly self-motivated with strong analytical skills, effective teamwork capabilities, and a persistent drive for innovation with relatively rich experience in scientific research.